

Orthogonal Annotation Benefits Barely-supervised Medical Image Segmentation

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Abstract

Recent trends in semi-supervised learning have significantly boosted the performance of 3D semi-supervised medical image segmentation. Compared with 2D images, 3D medical volumes involve information from different directions, e.g., transverse, sagittal, and coronal planes, so as to naturally provide complementary views. These complementary views and the intrinsic similarity among adjacent 3D slices inspire us to develop a novel annotation way and its corresponding semi-supervised model for effective segmentation. Specifically, we firstly propose the orthogonal annotation by only labeling two orthogonal slices in a labeled volume, which significantly relieves the burden of annotation. Then, we perform registration to obtain the initial pseudo labels for sparsely labeled volumes. Subsequently, by introducing unlabeled volumes, we propose a dual-network paradigm named Dense-Sparse Co-training (DeSCO) that exploits dense pseudo labels in early stage and sparse labels in later stage and meanwhile forces consistent output of two networks. Experimental results on three benchmark datasets validated our effectiveness in performance and efficiency in annotation. For example, with only 10 annotated slices, our method reaches a Dice up to 86.93% on KiTS19 dataset. Our code and models are available at <https://github.com/HengCai-NJU/DeSCO>.

1. Introduction

Medical image segmentation is one of the most critical vision tasks in medical image analysis field. Thanks to the development of deep learning-based methods [8, 11, 28, 32], segmentation performance has now been substantially improved. However, the current promising performance is at

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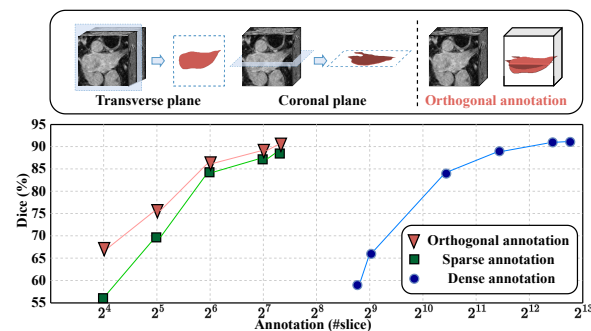


Figure 1. The upper figure illustrates our annotation method, each volume with annotations is labeled with only two orthogonal slices. The lower figure shows the comparison between the efficiency and effectiveness of our orthogonal annotation and other manners, including conventional dense annotation and previous sparse annotation which labels slices in one plane. All trained on LA [42] dataset with supervised setting. For sparse annotation and our orthogonal annotation, we train the models only on labeled voxels through partial cross-entropy and partial Dice loss.

the cost of large-scale manually precisely labeled dataset, which is prohibitively expensive and laborious to achieve. What's worse, different radiologists might provide different annotations even for a same image. Therefore, exploring ways to alleviate the requirement of quantity or quality of manual annotation is highly demanded. Mainstream methods typically follow two paradigms: 1) degrade annotation quality, i.e., weakly-supervised segmentation, and 2) reduce annotation quantity, i.e., semi-supervised segmentation.

Weakly-supervised segmentation methods usually utilize weak annotations, e.g., image-level label [16, 17], scribble [20, 21], point [3] or partial slices [5, 18]. Unfortunately, most of them are either difficult to distinguish some fuzzy boundaries or with additional large computational burden [15]. What's more, weakly-supervised setting usually requires coarse annotation for every single image. This is still a heavy burden for radiologists. Besides, most current methods originally developed for 2D segmentation could not directly utilize 3D volumetric information [9].

Different from these weakly-supervised methods, semi-supervised methods train segmentation models with a small amount of manually labeled data and a large amount of unlabeled data, which have achieved remarkable perfor-

正交标注助力弱监督医学图像分割

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摘要

近年来，半监督学习的发展显著提升了3D半监督医学图像分割的性能。与2D图像相比，3D医学体积包含来自不同方向的信息，例如横断面、矢状面和冠状面，从而自然地提供互补视图。这些互补视图以及相邻3D切片之间的内在相似性启发我们开发一种新颖的标注方式及其相应的半监督模型，以实现有效分割。具体而言，我们首先提出正交标注，即在一个标记体积中仅标注两个正交切片，这显著减轻了标注负担。然后，我们进行配准以获得稀疏标注体积的初始伪标签。随后，通过引入无标注体数据，我们提出了一种名为稠密-稀疏协同训练 (DeSCO) 的双网络范式，该范式在早期阶段利用稠密伪标签，在后期阶段利用稀疏标签，同时强制两个网络的输出保持一致。在三个基准数据集上的实验结果验证了我们在性能和标注效率方面的有效性。例如，仅使用10个标注切片，我们的方法在KiTS19数据集上达到了高达86.93%的Dice。我们的代码和模型可在

<https://github.com/HengCai-NJU/DeSCO>获取。

1. 引言

医学图像分割是医学图像分析领域中最关键的视觉任务之一。得益于基于深度学习的方法的发展 [8, 11, 28, 32], 分割性能现已得到大幅提升。然而，当前令人瞩目的性能是以

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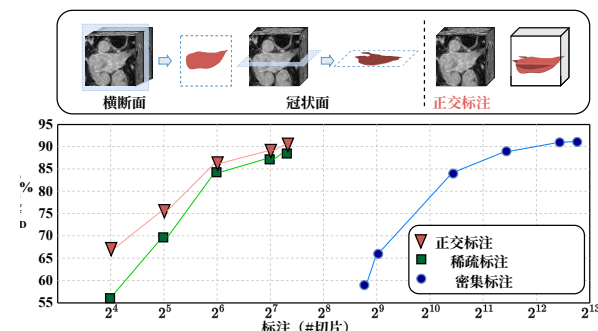


图1. 上图展示了我们的标注方法，每个带有标注的体积仅标注两个正交切片。下图比较了我们的正交标注与其他方式（包括传统的密集标注和以往在单个平面内标注切片的稀疏标注）的效率与效果。所有模型均在LA [42] 数据集上以监督设置进行训练。对于稀疏标注和我们的正交标注，我们仅通过部分交叉熵和部分Dice损失在标注体素上训练模型。

大规模人工精确标注数据集为代价换来的，而获取这样的数据集成本高昂且极为费力。更糟糕的是，即使是同一张图像，不同的放射科医生也可能提供不同的标注。因此，亟需探索减轻对人工标注数量或质量要求的方法。主流方法通常遵循两种范式：1) 降低标注质量，即弱监督分割；2) 减少标注数量，即半监督分割。

弱监督分割方法通常利用弱标注，例如图像级标签 [16, 17], 涂鸦 [20, 21], 点 [3] 或部分切片 [5, 18]。遗憾的是，这些方法大多要么难以区分某些模糊边界，要么带有额外的大量计算负担 [15]。此外，弱监督设置通常需要对每张单幅图像进行粗略标注。这对放射科医生来说仍然是沉重的负担。此外，大多数当前方法原本是为2D分割开发的，无法直接利用3D体数据信息 [9]。

与这些弱监督方法不同，半监督方法利用少量人工标注数据和大量未标注数据来训练分割模型，已取得了显著性

mance with an impressive deduction on demand for annotation [6, 19]. Despite their success, we notice most current semi-supervised segmentation methods still require full 3D annotation for each labeled volume. In fact, segmentation targets in adjacent slices of 3D volume are highly similar in both appearance and location, leading it redundant to label every slice. Although the sparse annotation is discussed in recent work [18], we notice these conventional methods still neglect the complementary views between different directions in 3D volume.

It is known that 3D medical volumes naturally contains different directions (*e.g.*, transverse, coronal planes) which provide complementary information from different views. And recent trends in semi-supervised learning [7, 40] have revealed that learning from complementary view is indeed beneficial. Thus, we wonder *whether a novel annotation method coupled with its corresponding model could be investigated by introducing this complementary relation into 3D semi-supervised medical image segmentation*.

In this paper, for labeled volume, we innovatively investigate a novel sparse annotation way—*orthogonal annotation, i.e.*, only to label two slices in its orthogonal direction (*e.g.*, transverse and coronal direction in Figure 1). We believe our annotation way has two merits: 1) it could largely force the model to learn from complementary views with two diversely initialized labeled slices, 2) it helps greatly reduce the label costs with fully utilizing the inter-slice similarity. Following very recent work [18], we name the setting as Barely-supervised Segmentation.

To incorporate our orthogonal annotation, the most intuitive thought about training strategy of a segmentation model is that only the voxels on the labeled slices contribute to the training. However, directly learning from this sparse annotation is unstable and the training is apt to collapse (shown in Sec. 4). Thus, we apply registration to spread supervision signals from slice to volume, where the result of label propagation can serve as the dense pseudo label for training. By performing registration, we obtain two sets of pseudo labels for volumes from orthogonal directions. Yet, the obtained pseudo labels are not promising enough to directly train a segmentation model using current existing semi-supervised methods, which is mainly due to the accumulation of error in the registration process.

Therefore, to leverage 1) the volumes with inaccurate pseudo labels and 2) the rest unlabeled volumes, we propose a simple yet effective end-to-end framework namely Dense-Sparse Co-training (DeSCO), which consists two segmentation models of a same structure. At the beginning of training, the models mainly learn from dense pseudo labels with a learning preference on voxels with more confident pseudo labels, *i.e.*, voxels near to registration source slice, and exploit unlabeled volumes through cross-supervision. After the models have been improved through training, we

gradually get rid of pseudo label until the supervised loss solely comes from sparse annotation. Meanwhile, the role of cross-supervision is gradually emphasized correspondingly. Because in the process of reaching consensus through cross-supervision, the mistake introduced by previous training on inaccurate pseudo labels could be revised. Overall, our contributions are three folds:

- A new annotation way that only labels two orthogonal slices for a labeled 3D volume, which greatly reduces the annotation burden.
- A novel barely-supervised 3D medical image segmentation framework to steadily utilize our high-efficient sparse annotation with coupled segmentation method.
- A dense-sparse co-training paradigm to learn from dense pseudo label and sparse label while leveraging unlabeled volumes to reduce noise by reaching consensus through cross-supervision.

Extensive experiments on three public datasets validate that our barely-supervised method is close to or even better than its upper bound, *i.e.*, semi-supervised methods with fully annotated labeled volumes. For example, on KiTS19, compared to Mean Teacher [36] that uses 320 labeled slices with a Dice of 84.98%, we only uses 10 labeled slices yet obtains a Dice of 86.93%.

2. Related Work

Semi-supervised Learning. Semi-supervised learning methods leverage a few labeled data and abundant unlabeled data to reduce the dependency on laborious annotation, and show their effectiveness on various tasks [22, 27, 45]. For example, Π -model [33] minimized the inconsistency between the outputs of differently disturbed inputs in order to improve the consistency of the model. Temporal Ensembling [33] and Mean Teacher [36] improved Π -model by leveraging the average of historic outputs to produce pseudo labels of higher quality. FixMatch [35] and FlexMatch [44] selected pseudo labels of higher quality with threshold. Some methods explored uncertainty by using novel consistency regularization, *e.g.*, UPS [31] and NP-Match [38]. The consistency regularization way used in these methods has greatly inspired current mainstream semi-supervised medical image segmentation methods.

Semi-supervised Medical Image Segmentation. To free radiologists from the burden of heavy annotation, many deep learning-based semi-supervised methods have been proposed for medical image segmentation in recent years.

Consistency regularization is a popular method to leverage unlabeled data. Li *et al.* [19] made a consistency constraint between the outputs of original images and transformed images. Similarly, Bortsova *et al.* [6] proposed a

能, 并显著降低了对标注的需求 [6, 19]。尽管这些方法取得了成功, 我们注意到当前大多数半监督分割方法仍然需要对每个标记体积进行完整的3D标注。事实上, 三维体数据中相邻切片的分割目标在外观和位置上高度相似, 导致对每个切片进行标注是冗余的。尽管近期工作讨论了稀疏标注 [18], 我们注意到这些传统方法仍然忽略了三维体数据中不同方向之间的互补视图。

众所周知, 3D医学体积天然包含不同方向(例如, 横断面、冠状面), 这些方向从不同视图提供互补信息。而半监督学习的最新趋势 [7, 40] 已表明, 从互补视图学习确实是有益的。因此, 我们想知道是否可以通过将这种互补关系引入3D半监督医学图像分割中, 来研究一种新的标注方法及其对应的模型。

本文针对标记体积, 创新性地研究了一种新的稀疏标注方式——**正交标注, 即仅在其正交方向上标注两个切片(例如图1中的横断面和冠状方向)**。我们相信我们的标注方式有两个优点: 1) 它可以在很大程度上迫使模型从两个多样化初始化的标注切片中学习互补视图, 2) 它通过充分利用切片间相似性, 极大地降低了标签成本。遵循最近的工作 [18], 我们将此设置命名为弱监督分割。

为了结合我们的正交标注, 训练分割模型最直观的策略是仅标注切片上的体素对训练有贡献。然而, 直接从这种稀疏标注中学习是不稳定的, 训练容易崩溃(见第4节)。因此, 我们采用配准将监督信号从切片传播到体积, 其中标签传播的结果可作为训练用的密集伪标签。通过执行配准, 我们从正交方向获得了两组体积的伪标签。然而, 所获得的伪标签不够理想, 无法直接使用现有的半监督方法来训练分割模型, 这主要是由于配准过程中的误差积累。

因此, 为了利用 1) 具有不准确伪标签的体积和 2) 其余未标注体积, 我们提出了一个简单而有效的端到端框架, 即稠密-稀疏协同训练(DeSCO), 它由两个具有相同结构的分割模型组成。在训练开始时, 模型主要从稠密伪标签中学习, 学习偏好集中在具有更高置信度伪标签的体素上, 即, 靠近配准源切片的体素, 并通过交叉监督利用未标注体积。在模型通过训练得到改进之后, 我们

逐渐摆脱伪标签, 直到监督损失仅来自稀疏标注。与此同时, 交叉监督的作用也相应地逐渐被强调。因为在通过交叉监督达成共识的过程中, 之前在不准确伪标签上训练所引入的错误可以得到修正。总体而言, 我们的贡献有三个方面:

- 一种新的标注方式, 仅对一个标注的三维体数据标注两个正交切片, 这大大减少了标注负担。
- 一种新颖的弱监督三维医学图像分割框架, 通过耦合分割方法稳定地利用我们的高效稀疏标注。
- 一种稠密-稀疏协同训练范式, 通过交叉监督达成共识, 从而利用稠密伪标签和稀疏标签进行学习, 同时借助无标注体数据来降低噪声。

在三个公开数据集上的大量实验验证了我们的弱监督方法接近甚至优于其上限, 即, 使用完全标注标记体积的半监督方法。例如, 在KiTS19上, 与使用320个标注切片、Dice为84.98%的Mean Teacher相比, [36] 我们仅使用10个标注切片, 却获得了86.93%的Dice。

2. 相关工作

半监督学习。半监督学习方法利用少量标注数据和大量未标注数据来减少对繁重标注的依赖, 并在各种任务上展现了其有效性 [22, 27, 45]。例如, Π -模型 [33]最小化不同扰动输入输出之间的不一致性, 以提高模型的一致性。Temporal Ensembling [33]和Mean Teacher [36] 通过利用历史输出的平均值来生成更高质量的伪标签, 改进了 Π -模型。FixMatch [35] 和FlexMatch [44]通过阈值选择更高质量的伪标签。一些方法通过使用新的一致性正则化来探索不确定性, 例如, UPS [31]和NP-Match [38]。这些方法中使用的一致性正则化方式极大地启发了当前主流的半监督医学图像分割方法。

半监督医学图像分割。为了将放射科医生从繁重的标注负担中解放出来, 近年来许多基于深度学习的半监督方法被提出用于医学图像分割。

一致性正则化是一种利用未标注数据的流行方法。Li 等人 [19]在原始图像和变换图像的输出之间施加了一致性约束。类似地, Bortsova 等人 [6] 提出了一种

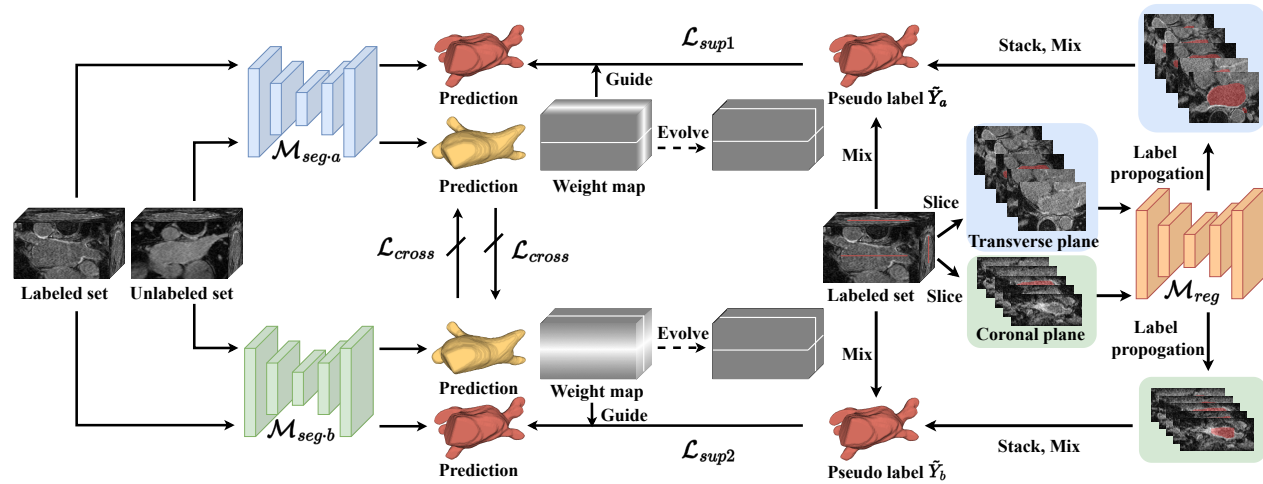


Figure 2. Overview of the proposed DeSCO paradigm. For a volume with orthogonal annotation, $\mathcal{M}_{reg}$ propagates the orthogonal annotation into the whole volume in two directions, and the results serve as pseudo labels to supervise segmentation model $\mathcal{M}_{seg-a}$ and $\mathcal{M}_{seg-b}$, respectively. For an unlabeled volume, $\mathcal{M}_{seg-a}$ and $\mathcal{M}_{seg-b}$ supervise each other with their outputs. Weight map guides $\mathcal{L}_{sup}$ and whiter areas mean higher weight.

method that learns to predict consistently under different transformations. Xie *et al.* [41] enforced the consistency of images in feature space. Different from previous work, Luo *et al.* [24] imposed the consistency constraint at task level. Shi *et al.* [34] incorporated consistency regularization and entropy minimization by treating certain/uncertain region with different training strategies. Besides, Co-training is another widely-used framework for semi-supervised medical image segmentation. Zhou *et al.* [46] applied multi-planar fusion to generate more accurate 3D pseudo label to train a co-training model. Xia *et al.* [40] introduced multi-view co-training into semi-supervised medical image segmentation and domain adaptation. And some works [2, 43] improving the quality of pseudo labels also showed great effectiveness. However, when applied to 3D segmentation tasks, as a heavy burden, most of them still require a dozen pixel-wise fully annotation slice-by-slice.

Weakly-supervised Segmentation. Weakly-supervised segmentation methods give pixel-level output with coarse annotation including image-level annotation [16, 17, 39], bounding box [12, 29], scribble [20, 21, 37] and even some points [3]. Among them, image-level annotation is the most popular setting and has been extensively studied. Many image-level weakly-supervised segmentation methods are based on class activation map (CAM). However, the area suggested by CAM is incapable of directly training a segmentation model as it only covers the most discriminative part of large objects and is liable to over-covers small objects. Other annotation methods suffer from the same problem more or less. The common problem in these annotation methods is that they discard the information of the boundaries between objects, which is crucial for segmentation tasks. Our annotation method takes into consideration

the precise delineation of objects and utilizes volumetric information of 3D medical images.

Remark. Our setting could be regarded as a variant of semi-supervised segmentation by requiring much fewer annotated slices. Our proposed orthogonal annotation has its merits in preserving disparity to boost the performance in semi-supervised scenario. Also, compared with weakly-supervised segmentation, our method only requires very few annotated slices while yielding promising results.

3. Method

In this work, we propose a novel annotation method called *orthogonal annotation* and a coupled training method comprised of 1) a registration module and 2) DesCO segmentation model. The training procedure is illustrated in Figure 2. We provide detailed introductions to the problem setting, our orthogonal annotation, registration module and the proposed DeSCO paradigm in the following parts.

3.1. Problem Setting and Notations

We aim to apply our orthogonal annotation in barely-supervised segmentation setting, *e.g.*, only 10 orthogonal slices from 5 volumes for an entire 3D medical image dataset. Training set contains N volumes $\{X_1, X_2, \dots, X_N\}$ of shape $H \times W \times D$, where H, W and D represent the height, width and depth of each volume, respectively. These volumes are further divided into $\{X_1, X_2, \dots, X_l\}$ with annotations and $\{X_{l+1}, X_{l+2}, \dots, X_N\}$ without annotations. Our methods annotates two slices from orthogonal planes denoted by a and b . The labeled slices of $X_i (1 \leq i \leq l)$ are denoted as X_{ia}^m and X_{ib}^n , which means m^{th} slice in plane a and n^{th} slice in plane b . Their annotations are denoted as Y_{ia}^m and Y_{ib}^n .

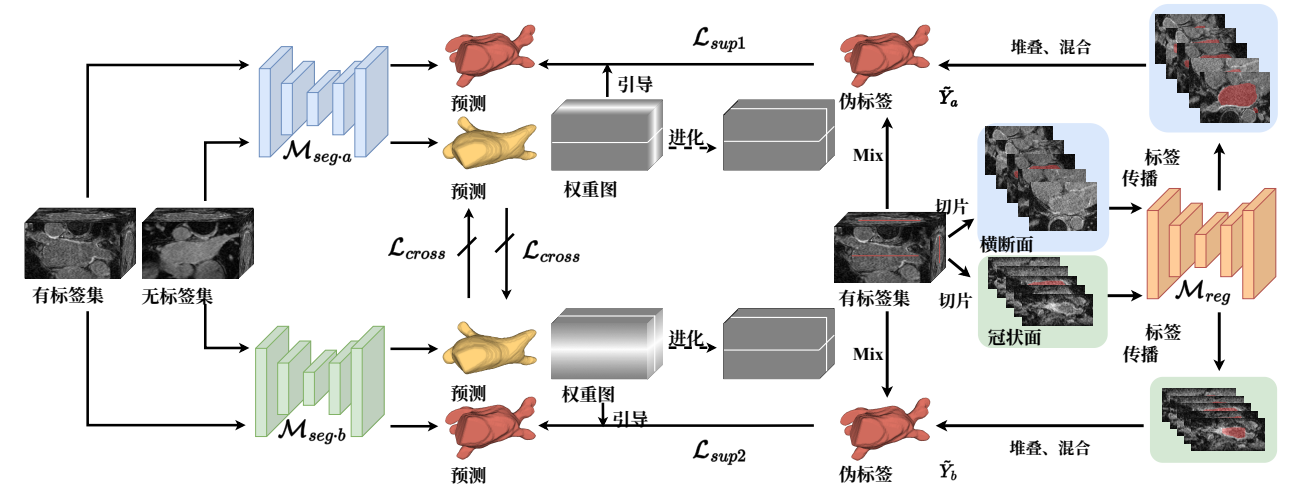


图 2. 所提出的 DeSCO 范式概览。对于具有正交标注的体积， $\mathcal{M}_{reg}$ 将正交标注沿两个方向传播至整个体积，其结果分别作为伪标签来监督分割模型 $\mathcal{M}_{seg-a}$ 和 $\mathcal{M}_{seg-b}$ 。对于无标注体积， $\mathcal{M}_{seg-a}$ 和 $\mathcal{M}_{seg-b}$ 利用各自的输出相互监督。权重图引导 $\mathcal{L}_{sup}$ ，较白的区域表示更高的权重。

方法，学习在不同变换下进行一致性预测。Xie 等人。[41]在特征空间中强制图像的一致性。与先前工作不同，Luo 等人。[24]在任务层面施加了一致性约束。Shi 等人。[34]通过以不同的训练策略处理确定/不确定区域，将一致性正则化和熵最小化相结合。此外，协同训练是另一种广泛用于半监督医学图像分割的框架。Zhou 等人。[46]应用多平面融合生成更准确的 3D 伪标签以训练协同训练模型。Xia 等人。[40]将多视角协同训练引入半监督医学图像分割和域适应。还有一些 [2, 43] 提升伪标签质量的工作也展现出了很好的效果。然而，当应用于 3D 分割任务时，作为一项沉重负担，大多数工作仍然需要大量逐像素的逐切片完全标注。

弱监督分割。弱监督分割方法利用粗略标注（包括图像级标注 [16, 17, 39], 边界框 [12, 29], 涂鸦 [20, 21, 37] 甚至一些点 [3]）给出像素级输出。其中，图像级标注是最流行的设置，并已被广泛研究。许多图像级弱监督分割方法基于类激活映射（CAM）。然而，CAM 所建议的区域无法直接训练分割模型，因为它仅覆盖大对象中最具判别性的部分，并且容易过度覆盖小对象。其他标注方法或多或少也面临同样的问题。这些标注方法的共同问题在于，它们丢弃了对象之间边界的信息，而这对分割任务至关重要。我们的标注方法考虑了

对象的精确描绘，并利用了 3D 医学图像的体积信息。

备注。我们的设置可被视为半监督分割的一种变体，其所需的标注切片数量要少得多。我们提出的正交标注在保留差异性以提升半监督场景下的性能方面具有其优势。此外，与弱监督分割相比，我们的方法仅需极少的标注切片即可获得令人满意的结果。

3. 方法

在这项工作中，我们提出了一种新颖的标注方法，称为正交标注，以及一种耦合训练方法，由 1) 配准模块和 2) DeSCO 分割模型组成。训练过程如图 2 所示。我们将在以下部分详细介绍问题设置、我们的正交标注、配准模块以及所提出的 DeSCO 范式。

3.1. 问题设置与符号说明

我们的目标是将正交标注应用于弱监督分割设置中，例如，整个三维医学图像数据集仅使用来自 5 个体积的 10 个正交切片。训练集包含 N 个体积 $\{X_1, X_2, \dots, X_N\}$ ，形状为 $H \times W \times D$ ，其中 H, W 和 D 分别表示每个体积的高度、宽度和深度。这些体积进一步分为 $\{X_1, X_2, \dots, X_l\}$ 个有标注和 $\{X_{l+1}, X_{l+2}, \dots, X_N\}$ 个无标注。我们的方法标注正交平面上的两个切片，分别用 a 和 b 表示。 $X_i (1 \leq i \leq l)$ 的标注切片表示为 X_{ia}^m 和 X_{ib}^n ，这意味着平面 a 中的 m^{th} 切片和平面 b 中的 n^{th} 切片。它们的标注表示为 Y_{ia}^m 和 Y_{ib}^n 。

3.2. Justification for Orthogonal Annotation

Our proposed orthogonal annotation (OA in short), which only labels two slices in orthogonal directions for a 3D volume, is highly efficient in annotation. Additionally, we now validate its effectiveness empirically and experimentally, which includes the discussion about 1) disparity, 2) feature representation, and 3) initial performance. We here take LA dataset [42] as an example, which contains 100 3D volumes detailed in Sec. 4.1.

OA preserves the disparity. As revealed by previous theoretical studies in semi-supervised learning [30], disparity (or disagreement) between views (or models) providing complementary information is the key to success. Now we explore the similarity of parallel slices and orthogonal slices from both image- and feature-level. The slices are chosen at random and the features are extracted from V-Net [28] trained with fully annotated data. We utilize the popularly-used Hilbert-Schmidt independence criterion (HSIC) [26] as the evaluation metric. Smaller value of HSIC, more independence of two variables. The results are shown in Figure 3, which reveal that parallel slices are more relevant than orthogonal slices in terms of both image and feature. It implies that parallel annotations have more redundancy of information. Besides, slices from different planes are inherently varied and complementary. Thus, for the same amount of annotation, orthogonal annotation could be superior to parallel annotation.

OA enhances consistency from different directions. We now visualize the features under different settings. Here are three models trained with only one labeled slice in each volume. The labeled slices are from transverse plane for models t_1 and t_2 and from coronal plane for model c . In other words, labeled slices used to train models t_1 and t_2 are parallel and those used to train models t_1 and c are orthogonal.

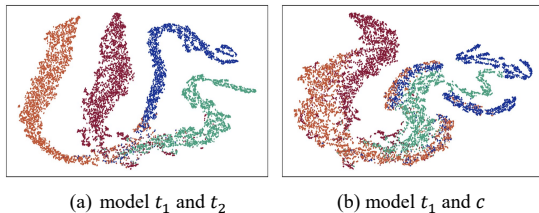


Figure 4. t-SNE result of feature for classification.

We extract the features before the classification layer of three models and use t-SNE to visualize them. The results are shown in Figure 4. Red and orange represent positive class while blue and aqua represent negative class. In par-

allel annotation, even the labeled slices come from a same plane, they learn different sub-patterns to distinguish foreground from background. Differently, the labeled slices to train t_1 and c share some common voxels (i.e., overlapping part) so that the features/predictions for the same class are closer. This result validates, by using orthogonal planes, different classes could have more consistent representation.

OA provides a promising initialization. To further explore the utility of the orthogonal annotation, we compare it to the parallel annotation in each plane. We trained models with three labeling manners: 1) one transverse plane slice and one coronal plane slice 2) two transverse plane slices 3) two coronal plane slices.

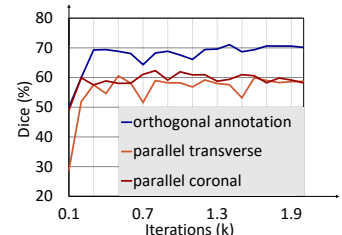


Figure 5. Dice coefficient on test set of models trained with different annotations.

performance greatly and this fact has always been neglected in medical image segmentation field.

These observations have revealed the superiority of our orthogonal annotation. More detailed settings of these experiments are provided in the supplementary material.

3.3. Registration Module

Registration seeks a spatial transformation to map an image to another image. For two slices: labeled slice (X_l, Y_l) and unlabeled slice X_u , registration obtains a spatial deformation field $\Phi(\cdot)$ by mapping from X_l to X_u . And pseudo label of X_u can be acquired by applying the same spatial deformation field to label Y_l , which can be formulated as $\hat{Y}_u = \Phi(Y_l)$.

As the segmentation targets in adjacent slices are usually similar in shape and size, it is feasible to propagate the label of a slice to its neighbouring slices through a registration module $\mathcal{M}_{reg}$. Note that registration module is an off-the-shelf tool that requires no training, so it introduces little extra computational cost except generating pseudo labels for only once.

For volume X with two orthogonal annotated slices (X_a^m, Y_a^m) and (X_b^n, Y_b^n) , we use the two slices to perform slice-by-slice label propagation and gradually generate two pseudo labels $\hat{Y}_a$ and $\hat{Y}_b$ for X through registration module $\mathcal{M}_{reg}$. First, we calculate spatial transformation fields that maps X_a^m to its closest slices X_a^{m-1} and X_a^{m+1} through $\mathcal{M}_{reg}$. Then the spatial transformation fields are applied to label Y_a^m to acquire pseudo labels $\hat{Y}_a^{m-1}$ and

3.2. 正交标注的合理性论证

我们提出的正交标注（简称OA），仅对三维体积的正交方向上的两个切片进行标注，在标注效率上非常高。此外，我们现在通过经验和实验验证其有效性，包括关于1) 差异性、2) 特征表示和3) 初始性能的讨论。我们这里以LA数据集 [42] 为例，该数据集包含100个三维体积，详见第4.1节。

正交标注保留了差异性。正如先前半监督学习中的理论研究所示，[30]，提供互补信息的视图（或模型）之间的差异性（或分歧）是成功的关键。现在我们分别从图像级和特征级探索平行切片与正交切片的相似性。切片随机选取，特征由V-Net [28] 在全标注数据上训练后提取。

我们采用广泛使用的希尔伯特-施密特独立性准则（HSIC）[26] 作为评估指标。HSIC值越小，两个变量的独立性越强。结果如图3所示，表明平行切片

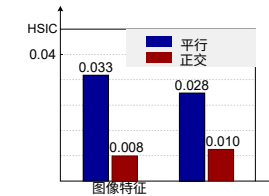


图3. 平行切片与正交切片的HSIC。

在图像和特征两个方面都比正交切片更具相关性。这意味着平行标注具有更多的信息冗余。此外，来自不同平面的切片本质上是多样且互补的。因此，在相同标注量的情况下，正交标注可能优于平行标注。

正交标注增强了不同方向之间的一致性。我们现在可视化不同设置下的特征。这里有三个模型，每个体积仅用一个标注切片训练。模型 t_1 和 t_2 的标注切片来自横断面，模型 c 的标注切片来自冠状面。换言之，用于训练模型 t_1 和 t_2 的标注切片是平行的，而用于训练模型 t_1 和 c 的标注切片是正交的。

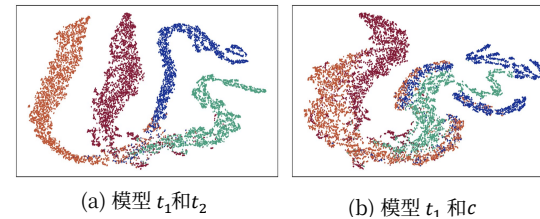


图 4. 用于分类的特征的 t-SNE 结果。

我们提取三个模型分类层之前的特征，并使用 t-SNE 对其进行可视化。结果如图4所示。红色和橙色代表正类，蓝色和水绿色代表负类。在平

行标注中，即使标注切片来自同一平面，它们也会学习不同的子模式来区分前景和背景。不同的是，标注切片

训练集 t_1 和 c 共享一些共同的体素（即重叠部分），从而使同一类别的特征/预测更加接近。该结果验证了，通过使用正交平面，不同类别可以具有更一致的表示。

正交标注提供了一种有前景的初始化。为了进一步探索正交标注的效用，我们将其与每个平面中的平行标注进行比较。我们使用三种标注方式训练模型：1) 一个横断面切片和一个冠状面切片 2) 两个横断面切片 3) 两个冠状面切片。

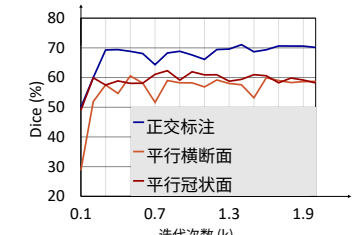


图5. 使用不同标注训练的模型在测试集上的Dice系数。

测试结果如图5所示。使用正交标注训练的模型在其初始阶段稳定且大幅优于其他两个模型。这表明，仅利用正交平面切片就能显著提升

性能，而这一事实在医学图像分割领域一直被忽视。

这些观察结果揭示了我们的正交标注的优越性。这些实验的更详细设置见补充材料。

3.3. 配准模块

配准旨在寻求一种空间变换，将一幅图像映射到另一幅图像。对于两个切片：标注切片 (X_l, Y_l) 和无标签切片 X_u ，配准通过从 X_l 到 X_u 的映射获得空间形变场 $\Phi(\cdot)$ 。而 X_u 的伪标签可以通过将相同的空间形变场应用于标签 Y_l 来获取，其可形式化为 $\hat{Y}_u = \Phi(Y_l)$ 。

由于相邻切片中的分割目标通常在形状和大小上相似，因此通过配准模块 $\mathcal{M}_{reg}$ 将一个切片的标签传播到其相邻切片是可行的。需要注意的是，配准模块是一种现成工具，无需训练，因此除了仅生成一次伪标签外，几乎不会引入额外的计算成本。

对于具有两个正交标注切片 (X_a^m, Y_a^m) 和 (X_b^n, Y_b^n) 的体积 X ，我们利用这两个切片通过配准模块 $\mathcal{M}_{reg}$ 对 X 执行逐切片标签传播，并逐步生成两个伪标签 $\hat{Y}_a$ 和 $\hat{Y}_b$ 。首先，我们计算通过 $\mathcal{M}_{reg}$ 将 X_a^m 映射到其最近切片 X_a^{m-1} 和 X_a^{m+1} 的空间变换场。然后，将空间变换场应用于标签 Y_a^m 以获取伪标签 $\hat{Y}_a^{m-1}$ 和

$\hat{Y}_a^{m+1}$, from which we further derive $\hat{Y}_a^{m-2}$ and $\hat{Y}_a^{m+2}$. Through the same procedure, we obtain pseudo labels for all slices viewed in plane a , and concatenate them to compose pseudo label $\hat{Y}_a$ for volume X . Similarly, the pseudo label $\hat{Y}_b$ can be easily obtained. So here we have pseudo label sets $\{\hat{Y}_{1a}, \hat{Y}_{2a}, \dots, \hat{Y}_{la}\}$ and $\{\hat{Y}_{1b}, \hat{Y}_{2b}, \dots, \hat{Y}_{lb}\}$ for those volumes with orthogonal annotation.

Performing label propagation suffers error accumulation as the pseudo label of each unlabeled slice is derived from its nearest pseudo label. Though error propagation can be greatly mitigated through morphology operations, it cannot be completely eliminated due to the limitation of registration itself. To alleviate the problem, we consider the importance of each slice in terms of their credibility.

3.4. Label Mixing

For each volume X with sparse orthogonal annotation, we have ground truth sparse annotation Y and dense registration pseudo label $\hat{Y}_a$ and $\hat{Y}_b$. Now we generate pseudo label $\tilde{Y}_a$ and $\tilde{Y}_b$ by mixing up dense registration pseudo label and ground truth sparse annotation:

$$\tilde{Y}_a = \text{LabelMix}(\hat{Y}_a, Y), \quad (1)$$

where $\text{LabelMix}(\cdot, \cdot)$ denotes a function mixing them by replacing the unlabeled voxels in Y with pseudo label in $\hat{Y}_a$. Similarly, $\tilde{Y}_b$ can be generated.

As mentioned in Sec. 3.3, different slices in $\hat{Y}$ should be considered differently according to the quality of pseudo labels. We assume the error rate of label propagation is a certain value, and the credibility of slice i can be denoted as $w_i = \alpha^d$, where d is the distance from slice i to registration source slice and α ($0 \leq \alpha < 1$) is the decay rate. On these grounds, we set the voxel-wise weight for each voxel in $\tilde{Y}_a$:

$$W_a^i = \begin{cases} 1 & \text{if voxel } i \text{ is labeled in } Y \\ \alpha^d & \text{otherwise.} \end{cases} \quad (2)$$

Similarly for the voxel weight map of $\tilde{Y}_b$, defined as W_b .

3.5. Dense-Sparse Co-Training

Our DeSCO paradigm consists of two 3D segmentation networks $\mathcal{M}_{seg-a}$ and $\mathcal{M}_{seg-b}$ of a same structure. It leverages labeled and unlabeled volumes simultaneously in a mini-batch. As mentioned above, every volume X_i ($i \leq l$) has two pseudo labels, $\hat{Y}_{ia}$ and $\hat{Y}_{ib}$. Segmentation network $\mathcal{M}_{seg-a}$ is trained with $\hat{Y}_{ia}$ and $\mathcal{M}_{seg-b}$ is trained with $\hat{Y}_{ib}$, respectively. In this way, the two segmentation networks mainly learn from two different perspectives with the targets obtained from different plane registration, and the disparity in orthogonal annotation is well-preserved. The supervised loss contains weighted cross-entropy loss and weighted dice

loss, which are formulated as:

$$\mathcal{L}_{ce} = -\frac{1}{\sum_{i=1}^{H \times W \times D} w_i} \sum_{i=1}^{H \times W \times D} w_i y_i \log p_i, \quad (3)$$

and

$$\mathcal{L}_{dice} = 1 - \frac{2 \times \sum_{i=1}^{H \times W \times D} w_i p_i y_i}{\sum_{i=1}^{H \times W \times D} w_i (p_i^2 + y_i^2)}, \quad (4)$$

where w_i is i^{th} voxel of weight map W and p_i, y_i denote the probability of foreground and pseudo label of voxel i respectively.

At the early stage of training, segmentation models need to learn from dense pseudo label for steady improvement. Segmentation networks are gradually improved during training and can produce better segmentation results than the initial pseudo labels. Therefore, the initial pseudo labels are actually becoming the obstacle for continual improvement. Based on this fact, we should weaken the influence of pseudo label. To implement this, we gradually decrease the decay rate α , and down to zero at last, which means the networks get rid of pseudo labels and only learn from sparse annotation.

The supervised loss is the weighted sum of cross-entropy loss and dice loss:

$$\mathcal{L}_{sup} = \frac{1}{2} \mathcal{L}_{ce} + \frac{1}{2} \mathcal{L}_{dice}. \quad (5)$$

For those volumes without annotations, the two segmentation models teach each other with their predictions. In a training iteration of volume X_i ($i \geq l+1$), $\mathcal{M}_{seg-a}$ takes X_i as input and generates its one-hot prediction, which will be partially selected as the pseudo label for $\mathcal{M}_{seg-b}$. Here we follow UAMT [43] by selecting those voxels whose uncertainty is lower than a threshold and produce mask M_{un} for better cross-supervision. Similarly, $\mathcal{M}_{seg-b}$ regards masked one-hot prediction of $\mathcal{M}_{seg-a}$ as the pseudo label. The loss is formulated as:

$$\mathcal{L}_{cross} = -\frac{1}{\sum_{i=1}^{H \times W \times D} m_i} \sum_{i=1}^{H \times W \times D} m_i \hat{y}_i \log p_i, \quad (6)$$

where m_i denotes the value in mask M_{un} , p_i and $\hat{y}_i$ denote the probability of foreground and one-hot pseudo label predicted by two models.

For the late stage of training, the supervised loss mainly comes from the labeled slices and the information have been learnt by segmentation models, so keeping supervised loss weight high is to no avail. Oppositely, the cross-supervision weight should be increased for the two networks can correct the mistakes from noisy pseudo label and reach a consensus through cross-supervision. Thus, the overall objective is the weighted sum of $\mathcal{L}_{sup}$ and $\mathcal{L}_{cross}$:

$$\mathcal{L} = (1 - \lambda) \mathcal{L}_{sup} + \lambda \mathcal{L}_{cross}, \quad (7)$$

$\hat{Y}_a^{m+1}$, 由此进一步推导出 $\hat{Y}_a^{m-2}$ 和 $\hat{Y}_a^{m+2}$ 。通过相同的流程，我们获取平面 a 中查看的所有切片的伪标签，并将它们拼接以组成体积 X 的伪标签 $\hat{Y}_a$ 。类似地，伪标签 $\hat{Y}_b$ 可以轻松获得。因此，这里我们为那些具有正交标注的体积准备了伪标签集 $\{\hat{Y}_{1a}, \hat{Y}_{2a}, \dots, \hat{Y}_{la}\}$ 和 $\{\hat{Y}_{1b}, \hat{Y}_{2b}, \dots, \hat{Y}_{lb}\}$ 。

执行标签传播会遭受误差累积，因为每个无标签切片的伪标签都源自其最近的伪标签。尽管通过形态学操作可以极大地缓解误差传播，但由于配准本身的局限性，它无法被完全消除。为了缓解这个问题，我们从可信度的角度考虑每个切片的重要性。

3.4. 标签混合

对于每个具有稀疏正交标注的体积 X ，我们有真实稀疏标注 Y 以及稠密配准伪标签 Y_a 和 Y_b 。现在我们通过混合稠密配准伪标签和真实稀疏标注来生成伪标签 $\tilde{Y}_a$ 和 $\tilde{Y}_b$ ：

$$\tilde{Y}_a = \text{LabelMix}(\hat{Y}_a, Y), \quad (1)$$

其中 $\text{LabelMix}(\cdot, \cdot)$ 表示一个函数，它通过将 Y 中的未标注体素替换为 Y_a 中的伪标签来混合它们。 $\tilde{Y}_a$ 类似地， $\tilde{Y}_b$ 可以生成。

如第3.3节所述， Y 中的不同切片应根据伪标签的质量被区别对待。我们假设标签传播的错误率是一个确定的值，切片 i 的可信度可以表示为 $w_i = \alpha^d$ ，其中 d 是切片 i 到配准源切片的距离， α ($0 \leq \alpha < 1$) 是衰减率。基于这些依据，我们为 Y_a 中的每个体素设置体素级权重：

$$W_a^i = \begin{cases} 1 & \text{if voxel } i \text{ is labeled in } Y \\ \alpha^d & \text{otherwise.} \end{cases} \quad (2)$$

类似地，对于 Y_b 的体素权重图，定义为 W_b 。

3.5. 密集-稀疏协同训练

我们的DeSCO范式由两个结构相同的3D分割网络 $\mathcal{M}_{seg-a}$ 和 $\mathcal{M}_{seg-b}$ 组成。它在一个小批量中同时利用标记体积和未标注体积。如上所述，每个体积 X_i ($i \leq l$) 有两个伪标签， Y_{ia} 和 Y_{ib} 。分割网络 $\mathcal{M}_{seg-a}$ 分别用 Y_{ia} 训练， $\mathcal{M}_{seg-b}$ 用 Y_{ib} 训练。通过这种方式，两个分割网络主要从两个不同的视角学习，目标来自不同的平面配准，正交标注中的差异性得到了很好的保留。监督损失包含加权交叉熵损失和加权Dice

损失，其公式为：

$$\mathcal{L}_{ce} = -\frac{1}{\sum_{i=1}^{H \times W \times D} w_i} \sum_{i=1}^{H \times W \times D} w_i y_i \log p_i, \quad (3)$$

and

$$\mathcal{L}_{dice} = 1 - \frac{2 \times \sum_{i=1}^{H \times W \times D} w_i p_i y_i}{\sum_{i=1}^{H \times W \times D} w_i (p_i^2 + y_i^2)}, \quad (4)$$

其中 w_i 是权重图 W 的 i^{th} 体素， p_i, y_i 分别表示体素 i 的前景概率和伪标签。

在训练的早期阶段，分割模型需要从密集伪标签中学习以稳步提升。分割网络在训练过程中逐渐改进，能够产生比初始伪标签更好的分割结果。因此，初始伪标签实际上正在成为持续改进的障碍。基于这一事实，我们应该削弱伪标签的影响。为了实现这一点，我们逐渐降低衰减率 α ，最终降至零，这意味着网络摆脱伪标签，仅从稀疏标注中学习。

监督损失是交叉熵损失和Dice损失的加权和：

$$\mathcal{L}_{sup} = \frac{1}{2} \mathcal{L}_{ce} + \frac{1}{2} \mathcal{L}_{dice}. \quad (5)$$

对于那些没有标注的体积，两个分割模型通过各自的预测互相教导。在体积 X_i ($i \geq l+1$) 的一次训练迭代中， $\mathcal{M}_{seg-a}$ 将 X_i 作为输入并生成其独热预测，该预测将被部分选为 $\mathcal{M}_{seg-b}$ 的伪标签。此处我们遵循 UAMT [43] 的方法，选择不确定性低于阈值的体素并生成掩码 M_{un} 以实现更好的交叉监督。类似地， $\mathcal{M}_{seg-b}$ 将 $\mathcal{M}_{seg-a}$ 的掩码独热预测视为伪标签。损失公式为：

$$\mathcal{L}_{cross} = -\frac{1}{\sum_{i=1}^{H \times W \times D} m_i} \sum_{i=1}^{H \times W \times D} m_i \hat{y}_i \log p_i, \quad (6)$$

其中 m_i 表示 mask M_{un} 中的值， p_i 和 $\hat{y}_i$ 分别表示两个模型预测的前景概率和独热伪标签。

在训练的后期阶段，监督损失主要来自有标签切片以及分割模型已学到的信息，因此保持较高的监督损失权重是徒劳的。相反，应增加交叉监督权重，以使两个网络能够纠正噪声伪标签中的错误，并通过交叉监督达成共识。因此，总体目标是 $\mathcal{L}_{sup}$ 和 $\mathcal{L}_{cross}$ 的加权和：

$$\mathcal{L} = (1 - \lambda) \mathcal{L}_{sup} + \lambda \mathcal{L}_{cross}, \quad (7)$$

where λ is a dynamic parameter gradually increasing to optimal cross-supervision weight λ_{oc} .

4. Experiments

4.1. Datasets

LA Dataset [42] contains 100 3D gadolinium-enhanced magnetic resonance images required using clinical whole-body MRI scanner and full annotations for left atrial cavity by radiologists. All scans have the same isotropic resolution of $0.625 \times 0.625 \times 0.625\text{mm}^3$ while their dimensions vary from each other.

KiTS19 Dataset [14] is a benchmark for kidney and kidney tumor segmentation. It contains 300 arterial phase abdominal CT where there are one or more kidney tumors and 210 of them have accessible kidney and kidney tumor delineations. The CTs have an average slice number of 216 and the slices thicknesses range from 1mm to 5mm. Most of slices in the transverse plane have the size of 512×512 .

LiTS Dataset [4] consists of 201 contrast-enhanced abdominal CT scans and corresponding segmentation of liver as well as liver tumor provided by various clinical sites. And 131 of segmentations are open to the public. There are high variations in pixel spacing and slice thickness.

4.2. Experiment Setting

For the selection of annotated slices, the principle is that the segmented target should be visible in the selected slice. For generalization, we simply select a fixed slice roughly located in the middle of the whole target.

For the implementation of our method, we adopt the SyNRA method in Ants [1] as our registration module and V-Net [28] as the backbone of our segmentation models.

For the training of segmentation models, we set the batch size as 2, one for labeled sample and one for unlabeled sample. We use Stochastic Gradient Descent (SGD) with momentum of 0.9 and weight decay of 0.0001 as optimizer. The learning rate is initialized as 0.01 and gradually decays to 0.0001. Training iteration is set to 6000 for all experiments. λ_{oc} is 0.8 for all three datasets. For the slice-wise weight decay rate, we initialize α to 0.95 and update every 1000 iterations according to cosine rampdown from [23].

For the evaluation and comparison of our method and other semi-supervised methods, we adopt four commonly used metrics, which are Dice coefficient, Jaccard coefficient, 95% Hausdorff Distance (95% HD) and Average Surface Distance (ASD). We compare our method with some classic and state-of-the-art (SOTA) methods, including Mean Teacher (MT) [36], Cross Pseudo Supervision (CPS) [10], Cross Teaching Between CNN and Transformer (CTBCT) [25] and Conservative-radical Network (CoraNet) [34]. All methods are implemented in 3D manners. MT, CPS, CoraNet and the CNN network of CTBCT

use V-Net [28] as backbone, the transformer network of CTBCT is implemented as UNETR, which is introduced in [13]. As those methods are not specifically designed for barely-supervised setting, for fairer comparisons, we designed two training settings for those methods. 1) Dense: we provided dense pseudo labels by registration as labeled training data. From the comparison between our method and other methods trained in this setting, we show that our method can better leverage the inaccurate pseudo labels while other methods suffers severe degradation. 2) Sparse: we provide original orthogonal labels without registration. And during training, the supervised loss only comes from labeled voxles. In the comparison between our method and other methods trained in this setting, we show that directly learning from few sparse annotations is almost unattainable, thus, generating initial pseudo labels is necessary and reasonable. All the experiments are implemented using PyTorch and an NVIDIA GeForce RTX 3090 GPU.

4.3. Compared with SOTA Methods

Results on LA Dataset. Following [43], we use 80 volumes for training and 20 for testing. 5 volumes of training data have orthogonal annotations on transverse plane and coronal plane, while the rest of the training data are used as unlabeled volumes. As there’s no validation set in the experiments, so we report the results at the end of 6000 iterations of training for all methods. We randomly crop the original size $112 \times 112 \times 88$ into $112 \times 112 \times 80$ for training, and the patch size remains the same during testing. The prediction is generated by sliding the patch window with a stride of 18 on the coronal and sagittal plane and 4 on the transverse plane. The comparison results between our method and other methods are presented as mean $\pm$ std of five cross-validation in Table 1. It is clear that our method outperforms other SOTA semi-supervised methods in all metrics except ASD. Compared to Mean Teacher trained with fully annotated volumes, our method uses only 2.3% annotations but there is only a 1.65% drop on Dice coefficient. We notice that these methods in Sparse setting work poorly, it is mainly because of the variety between slices and the lack of supervision signal on most parts of the whole volume. The segmentation result examples are shown in Figure 6.

Results on KiTS19 Dataset. We divided 210 samples into training set and test set, which have 190 and 20 samples respectively. Similar to LA, 5 volumes are randomly selected to label and the rest 185 volumes are unlabeled data. Patch size is set to $128 \times 128 \times 64$. The results are shown in Table 2. From the table we can see that our method outperforms all methods on all metrics. Note that our method even outperforms MT trained with 32 times the annotated slices. The segmentation result examples are shown in Figure 7.

Results on LiTS Dataset. For LiTS dataset, we also randomly select 5 volumes as labeled data, 95 volumes as

其中, λ 是一个动态参数, 逐渐增大至最优交叉监督权重 λ_{oco} .

4. 实验

4.1. 数据集

LA数据集 [42] 包含100例三维钆增强磁共振图像, 这些图像使用临床全身MRI扫描仪获取, 并由放射科医生提供左心腔的完整标注。所有扫描具有相同的各向同性分辨率 $0.625 \times 0.625 \times 0.625\text{毫米}^3$, 而它们的尺寸各不相同。

KiTS19数据集 [14]是肾脏和肾脏肿瘤分割的基准数据集。它包含300例动脉期腹部CT, 其中有一个或多个肾脏肿瘤, 且其中210例具有可获取的肾脏和肾脏肿瘤轮廓标注。这些CT的平均切片数为216, 层厚范围为1毫米至5毫米。横断面中的大多数切片尺寸为 512×512 。

LiTS数据集 [4] 包含201例对比增强腹部CT扫描, 以及由多个临床中心提供的肝脏和肝脏肿瘤的相应分割。其中131例分割已公开。像素间距和层厚存在较大差异。

4.2. 实验设置

对于标注切片的选择, 其原则是分割目标应在所选切片中可见。为便于泛化, 我们简单地选择一个大致位于整个目标中间的固定切片。

对于本文方法的实现, 我们采用Ants [1] 中的SyNRA方法作为配准模块, 并采用V-Net [28] 作为分割模型的骨干网络。

对于分割模型的训练, 我们将批大小设为2, 一个用于有标注样本, 一个用于无标注样本。我们使用动量为0.9、权重衰减为0.0001的随机梯度下降 (SGD) 作为优化器。学习率初始化为0.01, 并逐渐衰减至0.0001。所有实验的训练迭代次数均设为6000。 λ_{oc} 在三个数据集上均为0.8。对于逐切片权重衰减率, 我们将 α 初始化为0.95, 并根据 [23]的余弦衰减每1000次迭代更新一次。

为了评估和比较我们的方法与其他半监督方法, 我们采用了四种常用指标, 即Dice系数、Jaccard系数、95%豪斯多夫距离 (95% HD) 和平均表面距离 (ASD)。我们将我们的方法与一些经典和最先进 (SOTA) 的方法进行了比较, 包括Mean Teacher (MT) [36], 交叉伪监督 (CPS) [10], CNN与Transformer之间的交叉教学 (CTBCT) [25] 以及保守-激进网络 (CoraNet) [34]。所有方法均以3D方式实现。MT、CPS、CoraNet以及CTBCT的CNN网络

使用V-Net [28]作为主干网络, CTBCT的Transformer网络实现为UNETR, 其在 [13]中引入。由于这些方法并非专门为弱监督设置设计, 为了更公平的比较, 我们为这些方法设计了两种训练设置。1) 稠密: 我们通过配准提供稠密伪标签作为有标注训练数据。通过我们的方法与此设置下训练的其他方法的比较, 我们表明我们的方法能够更好地利用不准确的伪标签, 而其他方法则遭受严重退化。2) 稀疏: 我们提供原始的未配准正交标注。在训练过程中, 监督损失仅来自有标注体素。在我们的方法与此设置下训练的其他方法的比较中, 我们表明直接从少量稀疏标注中学习几乎是不可能实现的, 因此, 生成初始伪标签是必要且合理的。所有实验均使用PyTorch和NVIDIA GeForce RTX 3090 GPU实现。

4.3. 与SOTA方法比较

LA数据集上的结果。遵循 [43], 我们使用80个体积用于训练, 20个用于测试。训练数据中有5个体积在横断面和冠状面上具有正交标注, 其余训练数据用作未标注体积。由于实验中不存在验证集, 因此我们报告所有方法在6000次训练迭代结束时的结果。我们随机裁剪原始尺寸 $112 \times 112 \times 88$ 为 $112 \times 112 \times 80$ 用于训练, 测试期间图像块大小保持不变。预测是通过在冠状面和矢状面上以步长18、在横断面上以步长4滑动图像块窗口生成的。我们的方法与其他方法的比较结果以五次交叉验证的均值 $\pm$ 标准差形式展示在表1中。很明显, 除了平均表面距离外, 我们的方法在所有指标上均优于其他SOTA半监督方法。与使用全标注体积训练的Mean Teacher相比, 我们的方法仅使用2.3%的标注, 但Dice系数仅下降1.65%。我们注意到, 这些方法在稀疏设置下表现不佳, 这主要是因为切片之间的差异以及整个体积大部分区域缺乏监督信号。分割结果示例如图6所示。

KiTS19数据集上的结果。我们将210个样本分为训练集和测试集, 分别包含190和20个样本。与LA类似, 随机选择5个体积进行标注, 其余185个体积为未标注数据。图像块大小设置为 $128 \times 128 \times 64$ 。结果如表2所示。从表中可以看出, 我们的方法在所有指标上均优于所有方法。值得注意的是, 我们的方法甚至优于使用32倍标注切片训练的MT。分割结果示例如图7所示。

LiTS数据集上的结果。对于LiTS数据集, 我们还随机选择5个体积作为标注数据, 95个体积作为

Table 1. Comparison on LA dataset segmentation.

Method		Venue	Setting	Scans Used		Metrics			
				L / U Volumes	Labeled Slices	Dice (%)	Jaccard (%)	HD (voxel)	ASD (voxel)
Barely-supervised	MT [36]	NIPS' 17	Dense	5 / 75	10	74.41±4.07	60.87±4.35	27.87±3.96	5.65±1.12
			Sparse	5 / 75	10	63.17±2.28	47.48±2.38	39.87±4.32	14.75±2.41
	CPS [10]	CVPR'21	Dense	5 / 75	10	74.02±0.41	60.62±1.37	30.32±1.73	5.86±0.92
			Sparse	5 / 75	10	62.24±2.55	46.12±2.87	45.03±2.77	17.83±2.29
	CTBCT [25]	MIDL'22	Dense	5 / 75	10	73.12±5.50	59.37±5.80	26.51±2.47	5.38±1.25
			Sparse	5 / 75	10	64.71±1.49	48.89±1.14	42.09±2.13	15.91±0.83
	CoraNet [34]	TMI'22	Dense	5 / 75	10	40.63±1.34	31.94±2.41	40.88±3.73	20.19±1.57
			Sparse	5 / 75	10	51.16±3.63	38.45±3.30	44.70±1.08	18.46±0.78
	Ours	this paper	-	5 / 75	10	77.48±0.58	64.36±1.25	24.86±10.58	6.41±4.68
			-	5 / 75	10	77.48±0.58	64.36±1.25	24.86±10.58	6.41±4.68
Semi-supervised	MT [36]	NIPS' 17	-	5 / 75	440	79.13±2.37	66.92±2.82	17.41±1.85	4.76±1.36

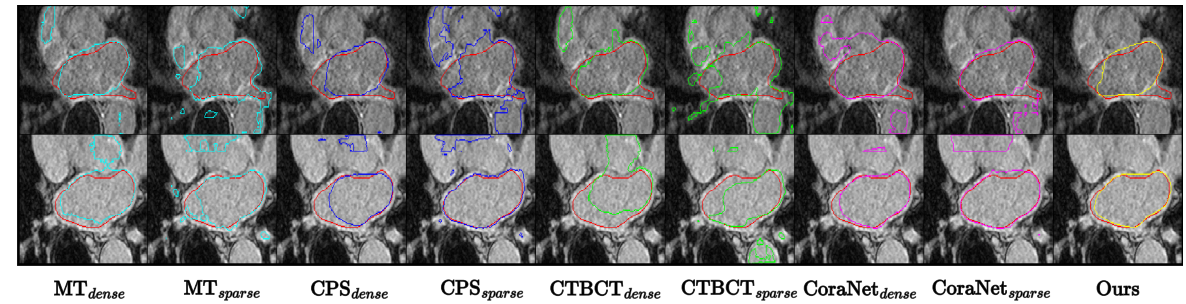


Figure 6. Visual segmentation examples from LA dataset. The red, cyan, blue, green, magenta, yellow curves denote the corresponding results of ground-truth, mean teacher [36], CPS [10], CTBCT [25], CoraNet [34] and ours, respectively.

Table 2. Comparison on KiTS19 dataset segmentation.

Method		Venue	Setting	Scans Used		Metrics			
				L / U Volumes	Labeled Slices	Dice (%)	Jaccard (%)	HD (voxel)	ASD (voxel)
Barely-supervised	MT [36]	NIPS' 17	Dense	5 / 185	10	78.16±1.99	66.96±1.41	17.15±5.93	5.40±2.25
			Sparse	5 / 185	10	32.16±13.78	20.41±11.04	67.42±12.73	32.35±9.83
	CPS [10]	CVPR'21	Dense	5 / 185	10	76.51±5.81	64.67±6.69	17.71±8.67	5.74±3.50
			Sparse	5 / 185	10	8.87±2.87	4.68±1.58	81.60±10.15	41.21±5.38
	CTBCT [25]	MIDL'22	Dense	5 / 185	10	80.65±0.83	69.99±2.27	14.88±6.07	4.28±1.88
			Sparse	5 / 185	10	17.87±0.87	9.88±0.55	76.06±8.57	37.83±5.17
	CoraNet [34]	TMI'22	Dense	5 / 185	10	43.08±28.61	33.27±23.87	24.17±3.88	8.61±2.11
			Sparse	5 / 185	10	28.64±25.05	18.90±19.44	72.62±22.57	35.22±13.43
	Ours	this paper	-	5 / 185	10	86.93±2.78	78.33±3.21	11.61±2.15	3.28±0.51
			-	5 / 185	10	86.93±2.78	78.33±3.21	11.61±2.15	3.28±0.51
Semi-supervised	MT [36]	NIPS' 17	-	5 / 185	320	84.98±4.68	76.51±5.63	16.51±0.18	4.61±0.83

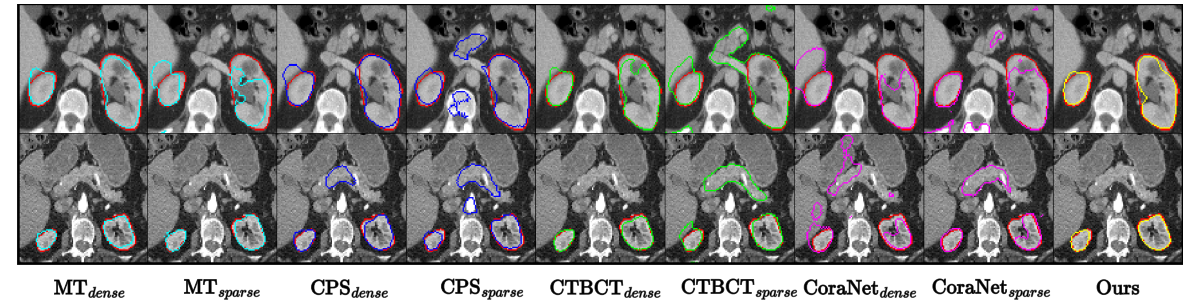


Figure 7. Visual segmentation examples from KiTS19 dataset. The red, cyan, blue, green, magenta, yellow curves denote the corresponding results of ground-truth, mean teacher [36], CPS [10], CTBCT [25], CoraNet [34] and ours, respectively.

unlabeled data and the rest 31 as testing data. We adopt random cropping to get patches of $176 \times 176 \times 64$ for training and testing. From Table 3, our method uses only 10 labeled slices and obtains a Dice coefficient close to 90%. Compared with other methods, our method outperforms all methods by a large margin. And our method is only 1.53% lower than MT trained with full volumetric annotation on Dice metric, but performs much better on HD and ASD.

The segmentation result examples are shown in Figure 8.

4.4. Ablation Study

Effectiveness of Each Component: In order to better understand and evaluate the components of our method, we conduct an ablation experiment on the KiTS19 dataset. Now we introduce each setting of our experiment: 1) training from mixed dense pseudo label (Dense), 2) leverag-

表 1. LA数据集分割对比。

方法		会议/期刊	设置	使用的扫描		指标			
				标注/伪标注 体积	标注切片	Dice (%)	Jaccard (%)	豪斯多夫距离 (体素)	ASD (体素)
弱监督	机器翻译 [36]	NIPS' 17	稠密	5 / 75	10	74.41±4.07	60.87±4.35	27.87±3.96	5.65±1.12
			稀疏	5 / 75	10	63.17±2.28	47.48±2.38	39.87±4.32	14.75±2.41
	CPS [10]	CVPR' 21	稠密	5 / 75	10	74.02±0.41	60.62±1.37	30.32±1.73	5.86±0.92
			稀疏	5 / 75	10	62.24±2.55	46.12±2.87	45.03±2.77	17.83±2.29
	CTBCT [25]	MIDL' 22	稠密	5 / 75	10	73.12±5.50	59.37±5.80	26.51±2.47	5.38±1.25
			稀疏	5 / 75	10	64.71±1.49	48.89±1.14	42.09±2.13	15.91±0.83
	CoraNet [34]	TMI' 22	稠密	5 / 75	10	40.63±1.34	31.94±2.41	40.88±3.73	20.19±1.57
			稀疏	5 / 75	10	51.16±3.63	38.45±3.30	44.70±1.08	18.46±0.78
	Ours	本文	-	5 / 75	10	77.48±0.58	64.36±1.25	24.86±10.58	6.41±4.68
			-	5 / 75	10	77.48±0.58	64.36±1.25	24.86±10.58	6.41±4.68
半监督	机器翻译 [36]	NIPS' 17	-	5 / 75	440	79.13±2.37	66.92±2.82	17.41±1.85	4.76±1.36

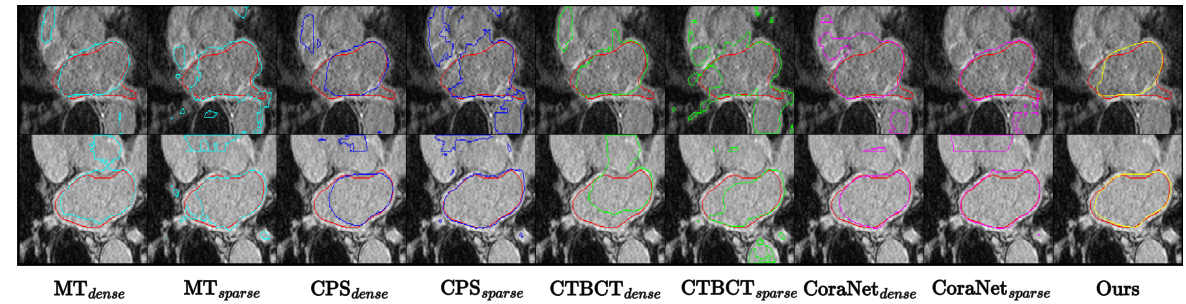


图6. LA数据集的视觉分割示例。红色、青色、蓝色、绿色、品红色、黄色曲线分别表示真实标签、Mean Teacher [36], CPS [10], CTBCT [25], CoraNet [34] 和本文方法的对应结果。

表2. KiTS19数据集分割的比较。

方法		会议/期刊	设置	使用的扫描		指标			
				标注/伪标注 体积	标注切片	Dice (%)	Jaccard (%)	HD (体素)	ASD (体素)
弱监督	机器翻译 [36]	NIPS' 17	稠密	5 / 185	10	78.16±1.99	66.96±1.41	17.15±5.93	5.40±2.25
			稀疏	5 / 185	10	32.16±13.78	20.41±11.04	67.42±12.73	32.35±9.83
	CPS [10]	CVPR' 21	稠密	5 / 185	10	76.51±5.81	64.67±6.69	17.71±8.67	5.74±3.50
			稀疏	5 / 185	10	8.87±2.87	4.68±1.58	81.60±10.15	41.21±5.38
	CTBCT [25]	MIDL' 22	稠密	5 / 185	10	80.65±0.83	69.99±2.27	14.88±6.07	4.28±1.88
			稀疏	5 / 185	10	17.87±0.87	9.88±0.55	76.06±8.57	37.83±5.17
	CoraNet [34]	TMI' 22	稠密	5 / 185	10	43.08±28.61	33.27±23.87	24.17±3.88	8.61±2.11
			稀疏	5 / 185	10	28.64±25.05	18.90±19.44	72.62±22.57	35.22±13.43
	Ours	本文	-	5 / 185	10	86.93±2.78	78.33±3.21	11.61±2.15	3.28±0.51
			-	5 / 185	10	86.93±2.78	78.33±3.21	11.61±2.15	3.28±0.51
半监督	机器翻译 [36]	NIPS' 17	-	5 / 185	320	84.98±4.68	76.51±5.63	16.51±0.18	4.61±0.83

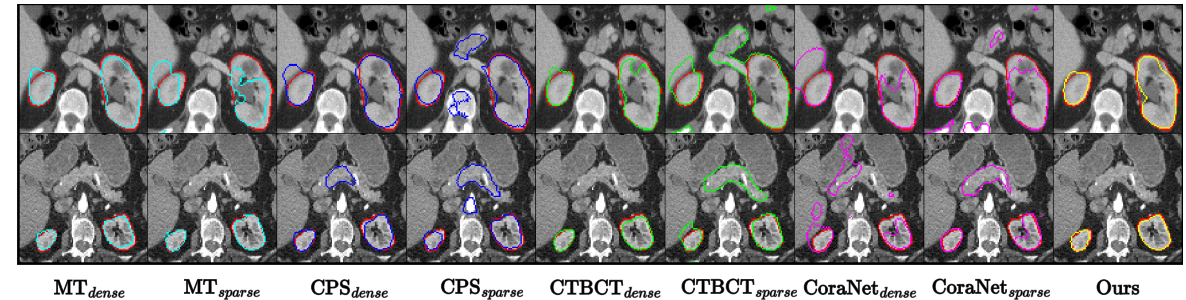


图7. 来自KiTS19数据集的视觉分割示例。红色、青色、蓝色、绿色、品红色、黄色曲线分别表示真实标签、Mean Teacher [36], CPS [10], CTBCT [25], CoraNet [34] 和本文方法的结果。

未标注数据，其余31个作为测试数据。我们采用随机裁剪来获取 $176 \times 176 \times 64$ 用于训练和测试的补丁。从表3中可以看出，我们的方法仅使用10个标注切片就获得了接近90%的Dice系数。与其他方法相比，我们的方法以较大优势优于所有方法。并且我们的方法在Dice度量上仅比使用完整体积标注训练的MT低1.53%，但在HD和ASD上表现好得多。

分割结果示例如图8所示。

4.4. 消融研究

各组件的有效性：为了更好地理解和评估我们方法的各个组件，我们在KiTS19数据集上进行了消融实验。现在我们介绍实验的每种设置：1) 从混合稠密伪标签（Dense）训练，2) 利用

Table 3. Comparison on LiTS dataset segmentation.

Method		Venue	Setting	Scans Used		Metrics			
				L / U Volumes	Labeled Slices	Dice (%)	Jaccard (%)	HD (voxel)	ASD (voxel)
Barely-supervised	MT [36]	NIPS'17	Dense	5 / 95	10	81.76±4.82	69.73±7.01	28.87±13.85	8.57±4.03
			Sparse	5 / 95	10	56.82±25.76	43.83±28.78	74.03±34.02	31.41±17.58
	CPS [10]	CVPR'21	Dense	5 / 95	10	73.86±10.13	59.73±13.16	32.78±14.68	10.73±5.20
			Sparse	5 / 95	10	20.46±2.15	11.48±1.35	92.27±3.03	41.34±2.09
	CTBCT [25]	MIDL'22	Dense	5 / 95	10	79.68±5.45	66.95±7.41	30.46±13.36	9.18±4.12
			Sparse	5 / 95	10	40.07±7.95	25.52±6.24	71.83±11.41	29.49±5.44
	CoraNet [34]	TMI'22	Dense	5 / 95	10	80.17±1.77	68.97±3.16	19.42±7.47	4.34±0.50
			Sparse	5 / 95	10	36.84±8.20	23.13±6.03	95.89±1.94	43.25±2.03
	Ours	this paper	Sparse	5 / 95	10	89.24±1.37	81.10±2.28	10.05±2.42	2.27±0.45
			-	5 / 95	10				
Semi-supervised	MT [36]	NIPS'17	-	5 / 95	320	90.77±1.91	83.62±3.11	18.32±7.17	4.95±1.81

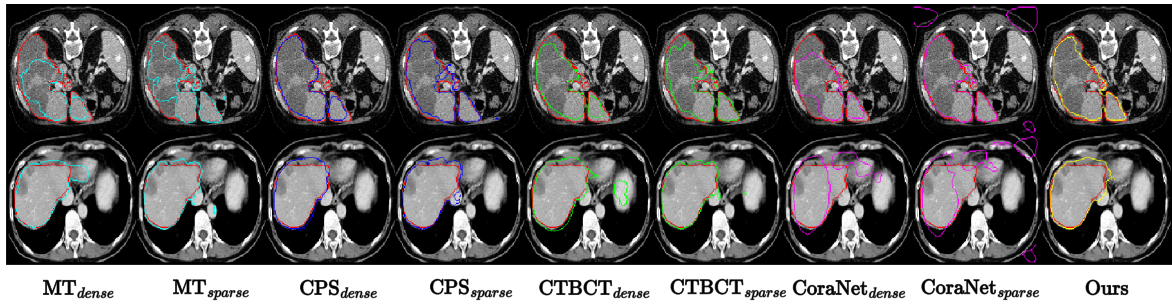


Figure 8. Visual segmentation examples from LiTS dataset. The red, cyan, blue, green, magenta, yellow curves denote the corresponding results of ground-truth, mean teacher [36], CPS [10], CTBCT [25], CoraNet [34] and ours, respectively.

Table 4. Ablation study on KiTS19 dataset.

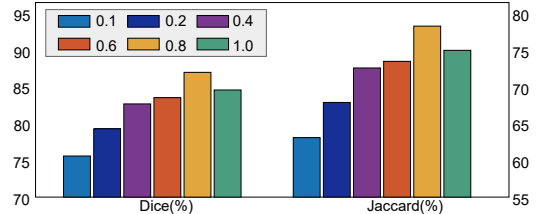
Methods	Dense	w_s	M_{un}	$D \rightarrow S$	$\lambda \uparrow$	Dice (%)	Jaccard (%)	HD (voxel)	ASD (voxel)
#1	✓					75.33±8.77	63.20±10.95	19.14±8.14	6.53±3.27
#2		✓				77.69±5.07	65.75±6.57	17.90±6.34	5.65±2.47
#3	✓	✓	✓			78.31±3.94	66.48±4.67	18.35±5.59	5.43±1.62
#4	✓	✓	✓	✓		78.45±6.45	67.03±9.13	30.50±6.28	9.57±3.32
#5	✓	✓	✓		✓	84.86±2.01	74.99±2.16	12.21±2.75	3.34±0.95
#6	✓	✓	✓	✓	✓	86.93±2.78	78.33±3.21	11.61±2.15	3.28±0.51

ing slice-wise weight (w_s), 3) uncertainty-guided cross-supervision (M_{un}), 4) gradually decreasing the weight decay rate α to zero, *i.e.*, learning from dense to sparse ($D \rightarrow S$), and 5) gradually increasing cross-supervision weight to λ_{oc} ($\lambda \uparrow$). The results are shown in Table 4.

The process of learning from dense to sparse and a high cross-supervision weight in the late stage play the most critical role. Throughout the training procedure, the performance of the segmentation models is improving, and they can produce better results than the initial pseudo labels, so the high weighted supervised loss is gradually hampering the segmentation networks from improving. Simply leveraging slice-wise weight w_s is also effective.

Quantitative Analysis on Hyper-Parameter: We also conduct exhaustive experiments of different optimal cross-supervision weights λ_{oc} on the KiTS19 datasets. Specifically, we set the optimal cross-supervision weight as 0.1, 0.2, 0.4, 0.6, 0.8 and 1.0. We find that setting λ_{oc} as 0.8 serves best. Detailed results are shown in Figure 9.

It is shown that optimal cross-supervision weight should not be too large or too small, because small λ_{oc} makes the networks concentrate on learning from registration labels and neglect the fact that the networks are improving during the training procedure, while too large λ_{oc} makes the net-

Figure 9. Segmentation result on KiTS dataset with different optimal cross-supervision weight λ_{oc} .

work gradually lose the supervision from the ground truth labels, and the performance tends to decrease through further cross-supervision without constraint.

5. Conclusion

In this paper, we proposed orthogonal annotation for 3D medical image segmentation which is to label two orthogonal slices for a volume, and we verified its high-efficiency. Then we applied this annotation manner in barely-supervised segmentation setting. To better leverage the volumes with orthogonal annotation and the large number of unlabeled volumes, we designed a dense-sparse co-training paradigm, which learns from dense pseudo label first, then reduces noise and gains further promotion from sparse annotation in later stage. Our method could better synthesize the knowledge from two planes through cross-supervision. Large quantities of experiments have validated the effectiveness of our orthogonal annotation and DeSCO paradigm, *e.g.*, our method could achieve a close or better performance compared with semi-supervised method, but with only 2%~3% annotations.

表 3. LiTS数据集分割对比。

方法		会议/期刊	设置	使用的扫描		指标			
				标注/未标注 体积	标注切片	Dice (%)	Jaccard (%)	豪斯多夫距离 (体素)	ASD (体素)
弱监督	机器翻译 [36]	NIPS' 17	稠密	5 / 95	10	81.76±4.82	69.73±7.01	28.87±13.85	8.57±4.03
			稀疏	5 / 95	10	56.82±25.76	43.83±28.78	74.03±34.02	31.41±17.58
	CPS [10]	CVPR' 21	稠密	5 / 95	10	73.86±10.13	59.73±13.16	32.78±14.68	10.73±5.20
			稀疏	5 / 95	10	20.46±2.15	11.48±1.35	92.27±3.03	41.34±2.09
	CTBCT [25]	MIDL' 22	稠密	5 / 95	10	79.68±5.45	66.95±7.41	30.46±13.36	9.18±4.12
			稀疏	5 / 95	10	40.07±7.95	25.52±6.24	71.83±11.41	29.49±5.44
	CoraNet [34]	TMI' 22	稠密	5 / 95	10	80.17±1.77	68.97±3.16	19.42±7.47	4.34±0.50
			稀疏	5 / 95	10	36.84±8.20	23.13±6.03	95.89±1.94	43.25±2.03
	Ours	本文	稠密	5 / 95	10	89.24±1.37	81.10±2.28	10.05±2.42	2.27±0.45
			稀疏	5 / 95	10				
半监督	机器翻译 [36]	NIPS' 17	-	5 / 95	320	90.77±1.91	83.62±3.11	18.32±7.17	4.95±1.81

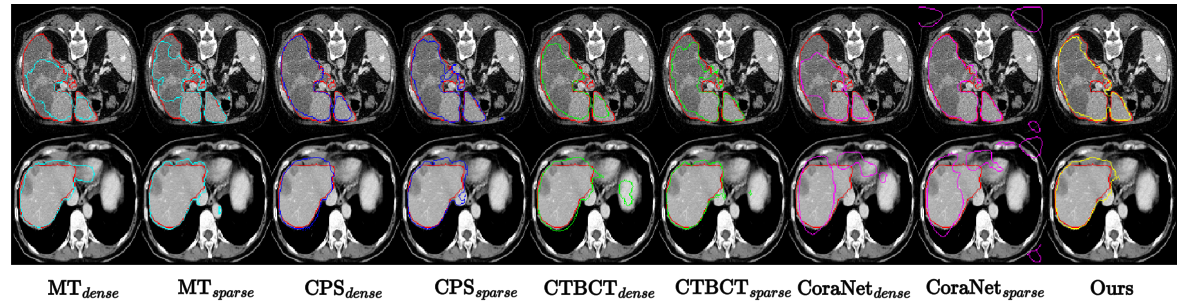


图8. LiTS数据集上的视觉分割示例。红色、青色、蓝色、绿色、品红色、黄色曲线分别表示真实标签、Mean Teacher [36],CPS [10], CTBCT [25],CoraNet [34]以及本文方法的结果。

表4. KiTS19数据集上的消融研究。

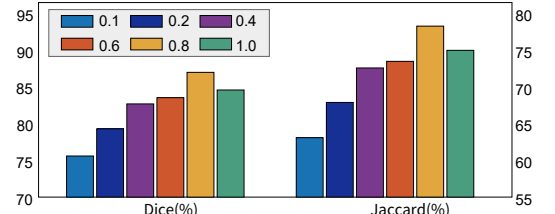
方法	稠密	w_s	M_{un}	$D \rightarrow S$	$\lambda \uparrow$	Dice (%)	Jaccard (%)	HD (体素)	ASD (体素)
#1	✓					75.33±8.77	63.20±10.95	19.14±8.14	6.53±3.27
#2		✓				77.69±5.07	65.75±6.57	17.90±6.34	5.65±2.47
#3	✓	✓	✓			78.31±3.94	66.48±4.67	18.35±5.59	5.43±1.62
#4	✓	✓	✓	✓		78.45±6.45	67.03±9.13	30.50±6.28	9.57±3.32
#5	✓	✓	✓		✓	84.86±2.01	74.99±2.16	12.21±2.75	3.34±0.95
#6	✓	✓	✓	✓	✓	86.93±2.78	78.33±3.21	11.61±2.15	3.28±0.51

切片级权重 (w_s) , 3) 不确定性引导的交叉监督 (M_{un}) , 4) 将权重衰减率逐渐降至 α 为零, 即, 从稠密到稀疏学习 ($D \rightarrow S$) , 以及5) 将交叉监督权重逐渐增加至 λ_{oc} ($\lambda \uparrow$) 。结果如表4所示。

从稠密到稀疏的学习过程以及后期较高的交叉监督权重起着最为关键的作用。在整个训练过程中, 分割模型的性能不断提升, 能够产生比初始伪标签更好的结果, 因此高权重的监督损失逐渐阻碍了分割网络的进一步改进。简单地利用切片级权重 w_s 也是有效的。

超参数的定量分析: 我们在KiTS19数据集上进行了不同最优交叉监督权重的详尽实验 λ_{oc} 。具体而言, 我们将最优交叉监督权重设置为0.1、0.2、0.4、0.6、0.8和1.0。我们发现将其设置为 λ_{oc} 0.8时效果最佳。详细结果如图9所示。

研究表明, 最优交叉监督权重既不应过大也不应过小, 因为过小的 λ_{oc} 会使网络集中于从配准标签中学习, 而忽视网络在训练过程中不断改进这一事实, 而过大的 λ_{oc} 则使网

图9. 在不同最优交叉监督权重 λ_{oc} 下KiTS数据集上的分割结果。

络逐渐失去来自真实标签的监督, 并且在无约束的进一步交叉监督下, 性能趋于下降。

5. 结论

在本文中, 我们提出了用于三维医学图像分割的正交标注方法, 即对一个体积标注两个正交切片, 并验证了其高效性。随后, 我们将这种标注方式应用于弱监督分割设置中。为了更好地利用具有正交标注的体积以及大量无标注体数据, 我们设计了一种密集-稀疏协同训练范式, 该范式首先从密集伪标签中学习, 然后在后期阶段减少噪声并从稀疏标注中获得进一步的提升。我们的方法能够通过交叉监督更好地综合来自两个平面的知识。大量实验验证了我们的正交标注和DeSCO范式的有效性, 例如, 与半监督方法相比, 我们的方法能够达到接近甚至更好的性能, 但仅需2%~3%的标注。

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